**Higher National Diploma in Engineering (Civil)**

**2nd Year, Second Semester Examination – 2019**

**CE 2212-Structural Design**

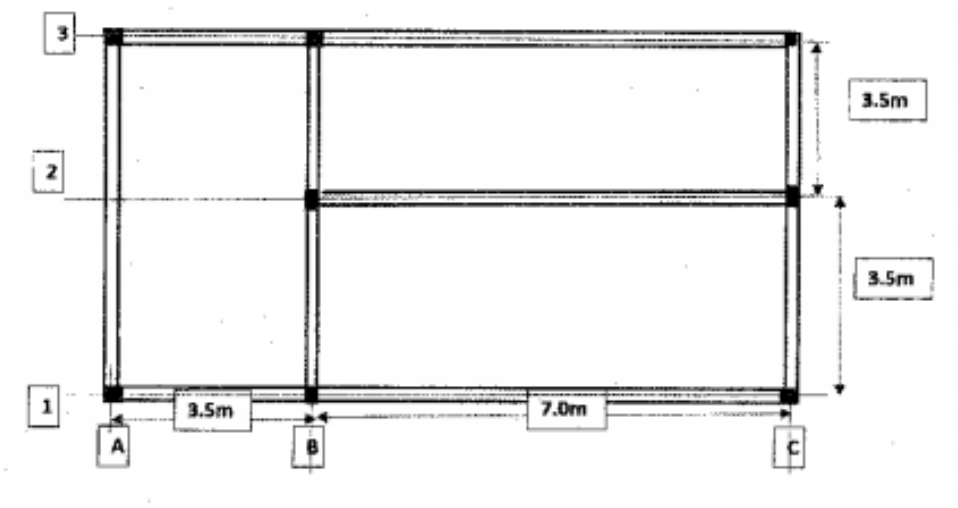


Figure 1

**The above drawing in figure 1 shows the slab, columns and beams arrangements of a two story building with roof slab. Plan and loading conditions are identical for both slabs. Q1, Q2 and Q3 are related to the Figure 1 and take following details for the calculations:**

- **Slab thickness = 175mm**
- **All Beam sizes = 300x600 mm**
- **Column size = 300x300 mm and Number of columns = 08**
- **Columns height = 4.0 m (both floors)** [13]
- **Load of Finishes of slabs = 1.5 kN/m$^2$ and imposed load = 4kN/m$^2$**
- **Density of Concrete = 24kN/m$^3$**
- **Characteristic strength of concrete, $f_{cu}$=35N/mm$^2$**

  **Main steel reinforcements, $f_y$=460N/mm$^2$ Shear links reinforcements, $f_{yv}$=250N/mm$^2$**

- **Exposure condition is mild and fire resistance of 1 hour.**

**Q1) Design the main reinforcement to slab BC23 in arrangement of figure 120. Perform the crack control and deflection checks also. Take the initial (Span/ Effective depth) ratio as 20, and all the slabs are simply supported on beams. Mention clearly the references of the code and assumptions made by you.**

**(Total 20 Marks)**

**Q2) Design the beam on Grid 2 from B to C shown in the above figure 1 and do the relevant checks25. Mention clearly the references of the code and assumptions made by you.**

**(Total 20 Marks)**

**Q3)**

**a) In the above figure 1, calculate the axial load on the column B1 at the ground floor level.**

**b) If the moment in the column is 30kNm, design the steel reinforcement to this column B1 at ground floor level29. Assume that Column is braced.**

**(Total 20 Marks)**

**Q4) A lap joint shown in Figure 2 which is a connection of two 10mm thick S 275 plates using two black bolts of Grade 8.8, 16mm diameter. There is one shear interface and it is assumed that the bolt is fully threaded.**

**a) Check the minimum and maximum edge and end distances.**

**b) Check the load capacity of the connection with respect to:**

**i. Bolt shear**

**ii. Bolt bearing**

**iii. Plate bearing**

**(Total 20 Marks)**

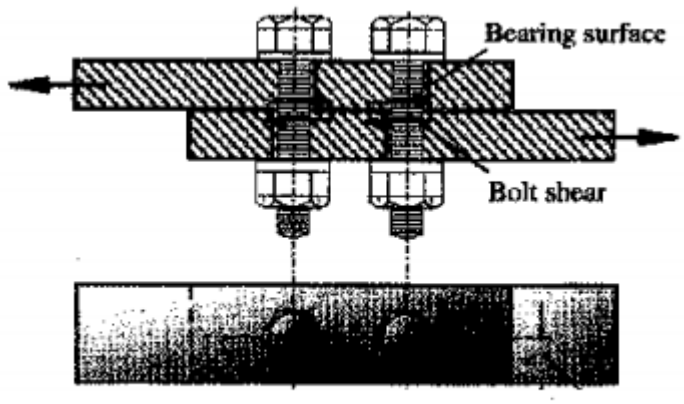


**Figure 2**

**Q5. A timber beam section of 75x260 mm with clear span 2.85m supports a uniformly distributed load of 10kN/m including self-weight of the beam40. The timber of strength class C16 Under service class 1. The beam is supported on two brick walls at the ends with the bearing length of 150mm and the ends of the beam are held in position and compression edge held in line42. Determine bending stress, shear stress, lateral buckling check and deflection of timber beam. Conclude whether the provided timber is adequate to resist the load.**

**(20 Marks)**

**Q6. Design the single leaf load-bearing wall shown in Figure 3 using mortar designation (iii) and either standard format bricks length 390mm, height 190mm and thickness 100mm. Assume the masonry units conform to Category I and the construction control category is 'normal'. Assume that:**

- **Effective height of the wall = 0.75 X clear height of the wall**

- **The partial factor of safety for normal manufacturing control and normal construction control, $ym = 3,5$**
- **Characteristic compressive strength of masonry, $fk = 2 \cdot 2N/mm^2$**
- **Neglect lateral load effects**
- **Design strength per unit length of wall =$\beta tfk/ym$**

**(20 Marks)**

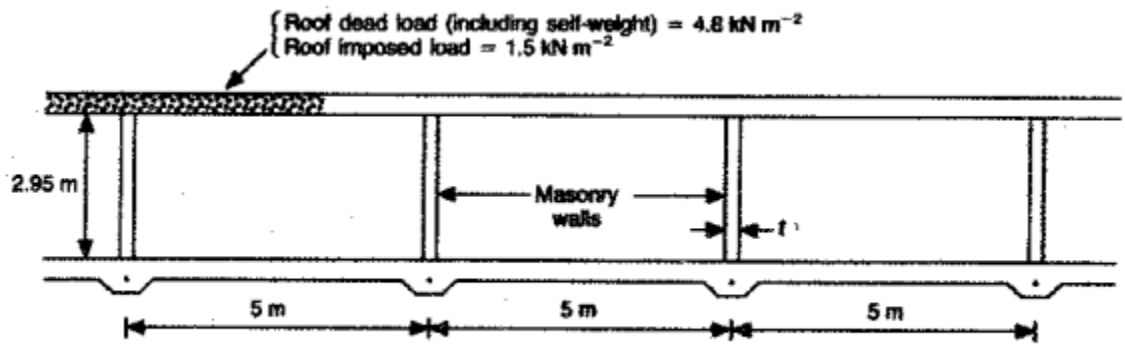


Figure 3

**Shear strength of bolts**

| Bolt Grade | Shear strength , $p_s$ (N/mm$^2$) |
|---|---|
| 4.6 | 160 |
| 8.8 | 375 |
| 10.9 | 400 |

**Bearing Strength of Bolts**

| Bolt Grade | Bearing Strength |
|---|---|
| 4.6 | 460 |
| 8.8 | 1000 |
| 10.9 | 1300 |

**Bearing Strength of Connected plates**

| Steel Grade | S275 | S355 | S460 |
|---|---|---|---|
| Bearing Strength(N/mm²) | 460 | 550 | 670 |

Bearing Capacity of bolt, $P_{bb} = dt_b p_{bb}$

Bearing capacity of connected plates, $P_{bs} = k_{bs} d t_p p_{bs}$ but $P_{bs} \leq 0.5 k_{bs} e t_p p_{bs}$

$k_{bs} = 1$ for Standard clearance holes

e = End distance

d = Nominal diameter of bolt

$t_p$ = Thickness of the connected plates

**Tensile stress area**

| Nominal size and tread diameter (mm) | Tensile stress area (mm²) |
|---|---|
| 15 | 84.5 |
| 16 | 157 |
| 20 | 245 |
| 22 | 303 |
| 24 | 353 |

**Area's of Reinforcement**

**Cross-sectional areas of groups of bars (mm²)**

| Bar Size | Number of Bars * Denotes non-preferred sizes | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 6mm | 28.3 | 56.5 | 84.8 | 113 | 141 | 170 | 198 | 226 | 255 | 283 |
| 8mm | 50.3 | 100 | 151 | 201 | 251 | 302 | 352 | 402 | 452 | 503 |
| 10mm | 78.5 | 157 | 236 | 314 | 339 | 471 | 550 | 628 | 707 | 785 |
| 12mm | 113.1 | 226 | 339 | 452 | 566 | 679 | 792 | 905 | 1718 | 1131 |
| 16mm | 201.1 | 402 | 603 | 804 | 1005 | 1206 | 1407 | 1608 | 1810 | 2011 |
| 20mm | 314.2 | 628 | 942 | 1257 | 1571 | 1885 | 2199 | 2513 | 2827 | 3142 |
| 25mm | 490.9 | 982 | 1473 | 1963 | 2454 | 2945 | 3436 | 3927 | 4418 | 4909 |
| 32mm | 804.2 | 1608 | 2413 | 3217 | 4021 | 4825 | 5630 | 6434 | 7238 | 8042 |
| 40mm | 1257.0 | 2513 | 3770 | 5026 | 6283 | 7540 | 8790 | 10053 | 11310 | 12566 |
| 50mm* | 1963.5 | 3927 | 5890 | 7854 | 9812 | 11781 | 13744 | 15708 | 17671 | 19635 |

**Cross-sectional areas for Specific Spacing in mm² / Metre Width**

| Bar Size | Spacing of Bars (Millimetres) * Denotes non-preferred sizes | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 75 | 100 | 125 | 150 | 175 | 200 | 225 | 250 | 275 | 300 |
| 6mm * | 377 | 283 | 226 | 188 | 162 | 141 | 125 | 113 | 103 | 94 |
| 8mm | 670 | 503 | 402 | 335 | 287 | 251 | 223 | 201 | 183 | 168 |
| 10mm | 1047 | 786 | 628 | 524 | 449 | 393 | 348 | 314 | 285 | 262 |
| 12mm | 1508 | 1131 | 905 | 754 | 646 | 565 | 503 | 452 | 411 | 377 |
| 16mm | 2681 | 2011 | 1608 | 1340 | 1149 | 1005 | 894 | 801 | 731 | 670 |
| 20mm | 4189 | 3142 | 2513 | 2094 | 1795 | 1571 | 1396 | 1257 | 1143 | 1047 |
| 25mm | 6545 | 4909 | 3927 | 3272 | 2805 | 2454 | 2182 | 1963 | 1785 | 1636 |
| 32mm | - | 8042 | 6434 | 5362 | 4596 | 4021 | 3574 | 3217 | 2924 | 2681 |
| 40mm | - | - | 10053 | 8378 | 7181 | 6283 | 5587 | 5027 | 4571 | 4189 |
| 50mm* | - | - | - | 13090 | 11220 | 9617 | 8727 | 7854 | 7140 | 6545 |

**Table 7 — Capacity reduction factor, $\beta$**

| Slenderness ratio $h_{ef}/t_{ef}$ | Eccentricity at top of wall, $e_x$ | | | |
|---|---|---|---|---|
| | Up to $0.05t$ (see note 1) | $0.1t$ | $0.2t$ | $0.3t$ |
| 0 | 1.00 | 0.88 | 0.66 | 0.44 |
| 6 | 1.00 | 0.88 | 0.66 | 0.44 |
| 8 | 1.00 | 0.88 | 0.66 | 0.44 |
| 10 | 0.97 | 0.88 | 0.66 | 0.44 |
| 12 | 0.93 | 0.87 | 0.66 | 0.44 |
| 14 | 0.89 | 0.83 | 0.66 | 0.44 |
| 16 | 0.83 | 0.77 | 0.64 | 0.44 |
| 18 | 0.77 | 0.70 | 0.57 | 0.44 |
| 20 | 0.70 | 0.64 | 0.51 | 0.37 |
| 22 | 0.62 | 0.56 | 0.43 | 0.30 |
| 24 | 0.53 | 0.47 | 0.34 | |
| 26 | 0.45 | 0.38 | | |
| 27 | 0.40 | 0.33 | | |

NOTE 1 It is not necessary to consider the effects of eccentricities up to and including $0.05t$.

NOTE 2 Linear interpolation between eccentricities and slenderness ratios is permitted.

NOTE 3 The derivation of $\beta$ is given in Appendix B.

**Table 2 — Characteristic compressive strength of masonry, $f_k$, in N/mm²**

| a) Constructed with standard format bricks | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Mortar designation | Compressive strength of unit (N/mm²) | | | | | | | | |
| | 5 | 10 | 15 | 20 | 27.5 | 35 | 50 | 70 | 100 |
| (i) | 2.5 | 4.4 | 6.0 | 7.4 | 9.2 | 11.4 | 15.0 | 19.2 | 24.0 |
| (ii) | 2.5 | 4.2 | 5.3 | 6.4 | 7.9 | 9.4 | 12.2 | 15.1 | 18.2 |
| (iii) | 2.5 | 4.1 | 5.0 | 5.8 | 7.1 | 8.5 | 10.6 | 13.1 | 15.5 |
| (iv) | 2.2 | 3.5 | 4.4 | 5.2 | 6.2 | 7.3 | 9.0 | 10.8 | 12.7 |



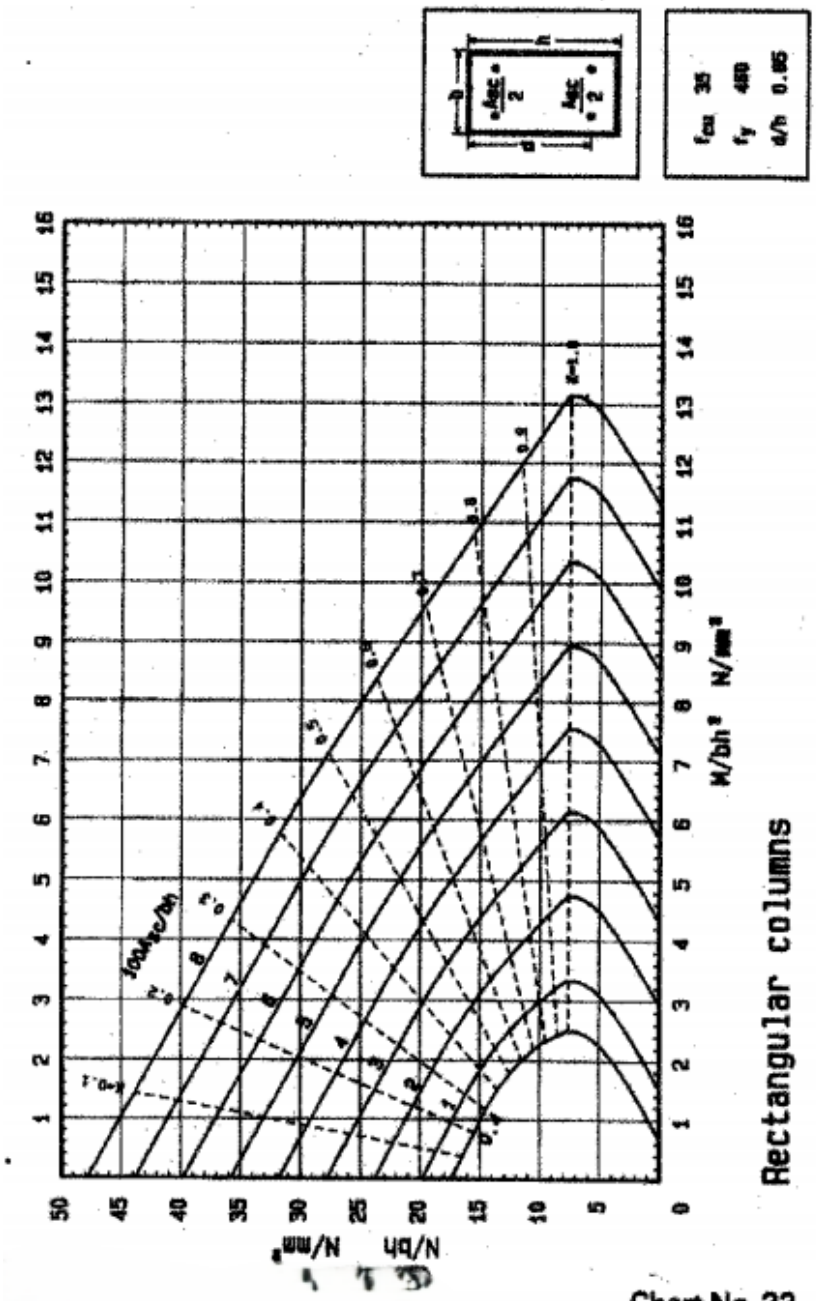


Chart No. 33